

Data Driven methods

① why do we need data driven methods ?

② classification vs Regression ?

③ what are the metrics for classification ?

a) accuracy

b) precision

c) Recall

d) F1 score

f) ROC - AUC

g) confusion matrix

④ True positive $\Rightarrow$ model correctly predicts positive class.

⊗ False positive $\Rightarrow$ model incorrectly predicts a positive class.

⊗ True negative $\Rightarrow$ model correctly predicts a negative class

⊗ False negative $\Rightarrow$ model incorrectly predicts a negative class.

$$\text{accuracy} = \frac{\text{True positives} + \text{True negatives}}{\text{True PS} + \text{True neg} + \text{False pos} + \text{False negative}}$$

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

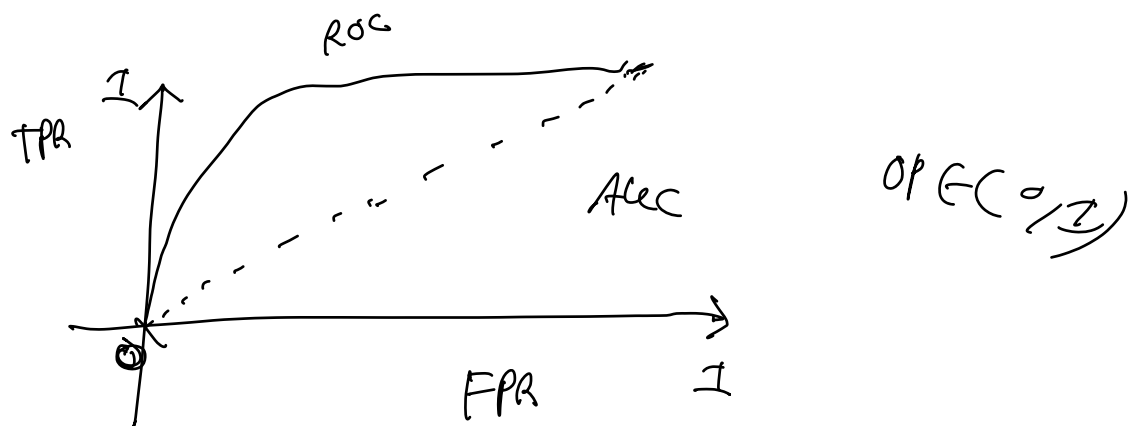
$$F1 \text{ score} = 2 \left[\frac{\text{precision} \times \text{Recall}}{\text{precision} + \text{Recall}} \right]$$

Roc $\rightarrow$ Receiver operating characteris.

$$TPR (\text{Recall}) = \frac{TP}{TP + FN} \quad / \quad \text{Specificity} =$$

$$\frac{TN}{TN + FP} \quad /$$

$$FPR = 1 - \left(\frac{TN}{TN + FP} \right) = \frac{FP}{TN + FP}$$



confusion matrix

		Actual	
		+	-
pred	+	(TP)	(FP)
	-	(FN)	(TN)

⊗ EER (TPR = FPR)

(false acceptance rate =
false reject rate)

Metrics for regression :-

a) $MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$

b) $RMSE = \sqrt{MSE}$

c) $MAE = \frac{\sum |y_i - \hat{y}_i|}{n}$

⊗ supervised learning, unsupervised learning.

⊗ Discriminative model, generative model
 $P(y|x)$ $P(y, x)$

⊗ Linear regression:-

$$y = \theta_1 x_1 + \theta_2 x_2 \dots + \theta_0$$

↳ + threshold fn $\rightarrow$
linear classifier / perceptron

⊗ sigmoid = $\frac{1}{1 + e^{-\sigma u}}$,
Domain $(-\infty, \infty)$

Range (0, 1)

⊗ softmax = normalized sigmoid ⇒

$$\frac{e^{x_i}}{\sum_{j=1}^k x_j}$$

$$x^{\rightarrow} = (3, 2, 0).$$

$$\Rightarrow e^3, e^2, e^0$$

$$\text{Denominator} = e^3 + e^2 + e^0$$

$$\text{softmax}_1 = \frac{e^3}{e^3 + e^2 + e^0} /$$

$$\frac{e^2}{e^3 + e^2 + e^0} /$$

$$\frac{e^0}{e^1 + e^2 + e^0}$$

cost fn or loss fn.

$$y = \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_0$$

$$\Rightarrow y = \theta_1 x_1 + \theta_0$$

$$(\theta_1 = 9, \theta_0 = 7)$$

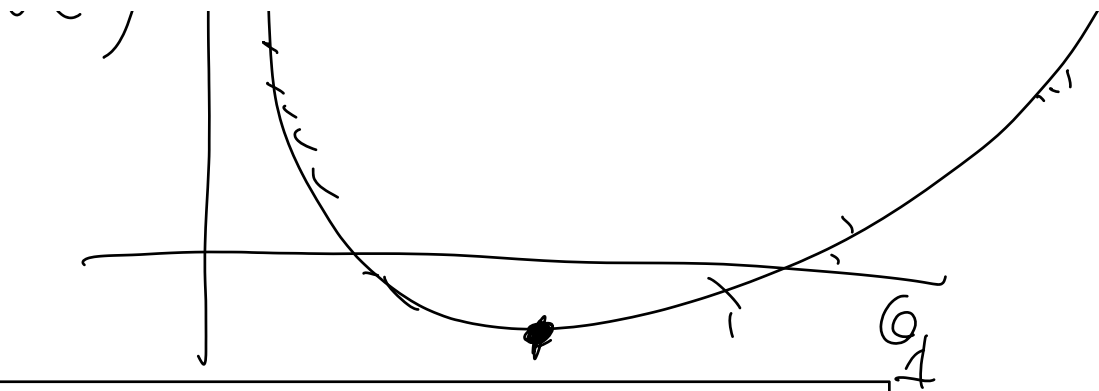
$(X, y) \rightarrow$ training set.

$$\Rightarrow \min_{\theta_0, \theta_1} \sum (\theta_1 x_1 + \theta_0 - y)^2 \frac{1}{2N} = J(\theta_0, \theta_1)$$

(loss) for linear regression.

Gradient descent:

$$J(\theta_1) \quad | \quad |$$



$$\hat{\theta}_j = \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

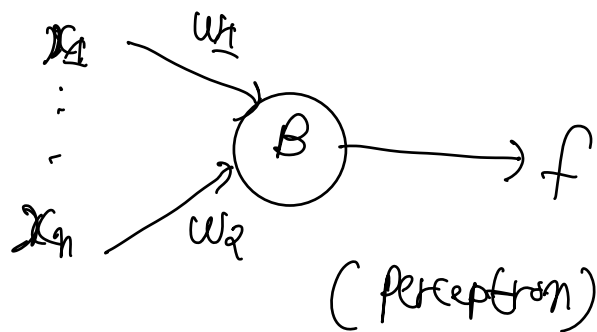
, $j = 0, 1$

α = learning rate.
(step size)

repeat until convergence.

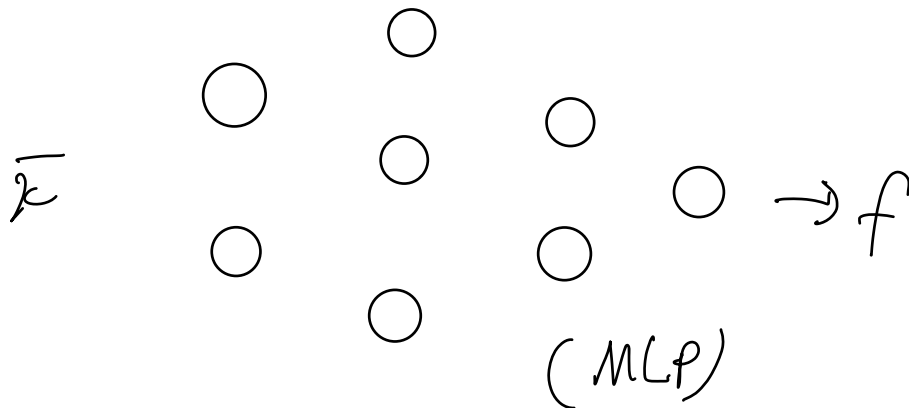
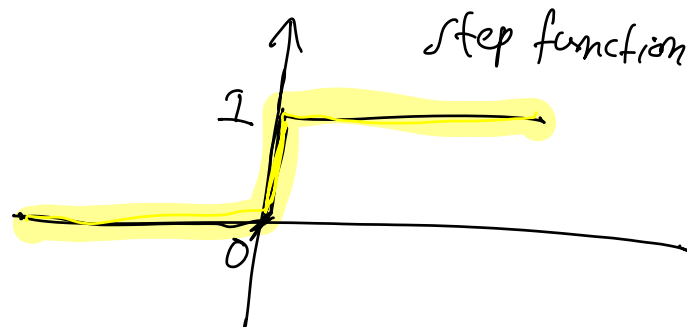
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Neural Networks



$\bar{x}\bar{w} + b \leq 0 \rightarrow \text{No fire}$	0
$\bar{x}\bar{w} + b > 0 \rightarrow \text{fire}$	1

(Linear classifier, NAND)



Convolutional Neural Networks

$\otimes (n \times n) * (f \times f)$

$$= (n - f + 1) \times (n - f + 1)$$

⊗

$$= (n - f + 1 + 2p) \times (n - f + 1 + 2p)$$

padding

$$= \left(\frac{(n + 2p - f)}{s} + 1 \right) \times$$

$$\left(\frac{(n + 2p - f)}{s} + 1 \right)$$

stride